Big Data for Official Statistics

COMMUTING TIME AND TRANSPORTATION MODES IN ITALY: A TERRITORIAL AND TEMPORAL ANALYSIS (2001–2024)

Insights from ISTAT Mobility Data and Machine Learning on Urban and Regional Dynamics





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Data & Methodology Overview

• Data Source:
ISTAT mobility data (**Spostamenti per andare a lavoro - regioni e tipo di comune**);
ISTAT Income data (**Reddito disponibile delle famiglie nelle regioni italiane**)

• Time Period Covered:
2001 to 2024, though it is important to note that data for 2004 is explicitly missing. Regional income data is available only up to 2023. Data are taken in real observed values (THV), not percentages (HSC)

Data Preprocessing



Loading and cleaning raw ISTAT data, indexing by Territorio (Territory) and TIME_PERIOD (Year).

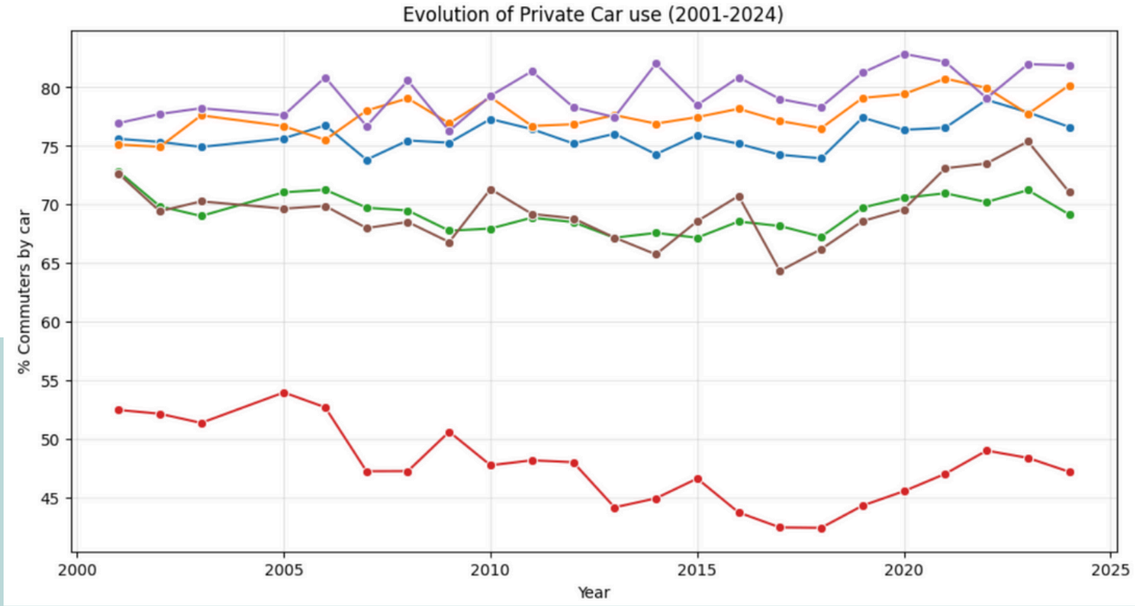


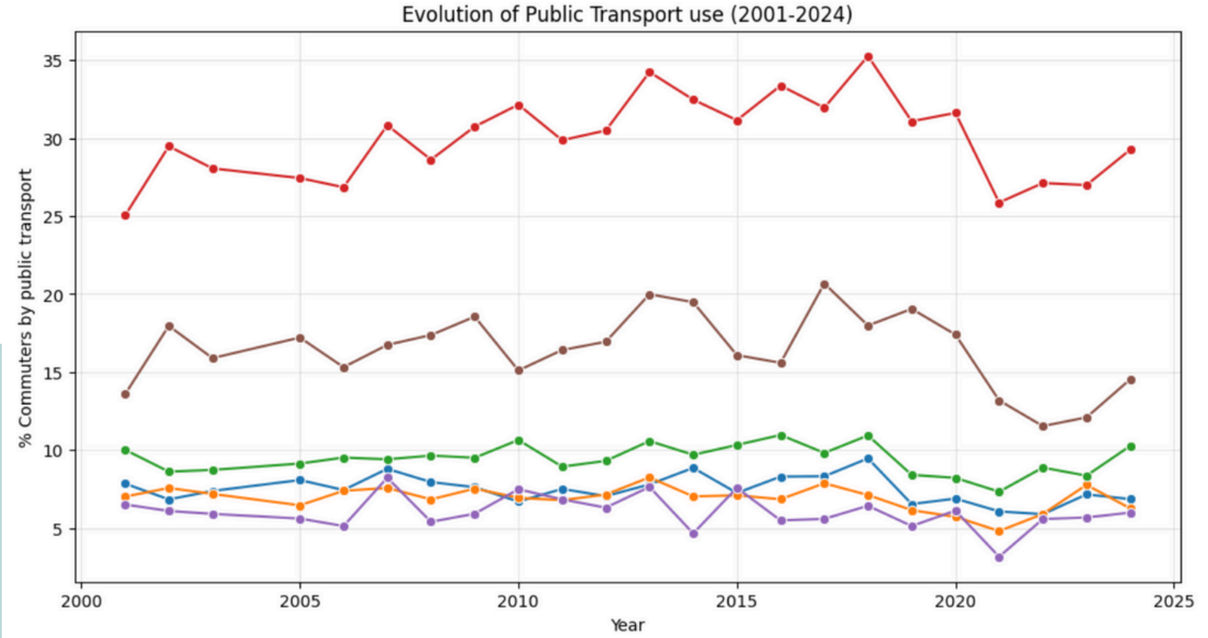
Calculating aggregates like total observations, missing time bracket (16-30 minutes) and percentages classes of means of transport (Car, Public, Motorbike, Feet/Bike)



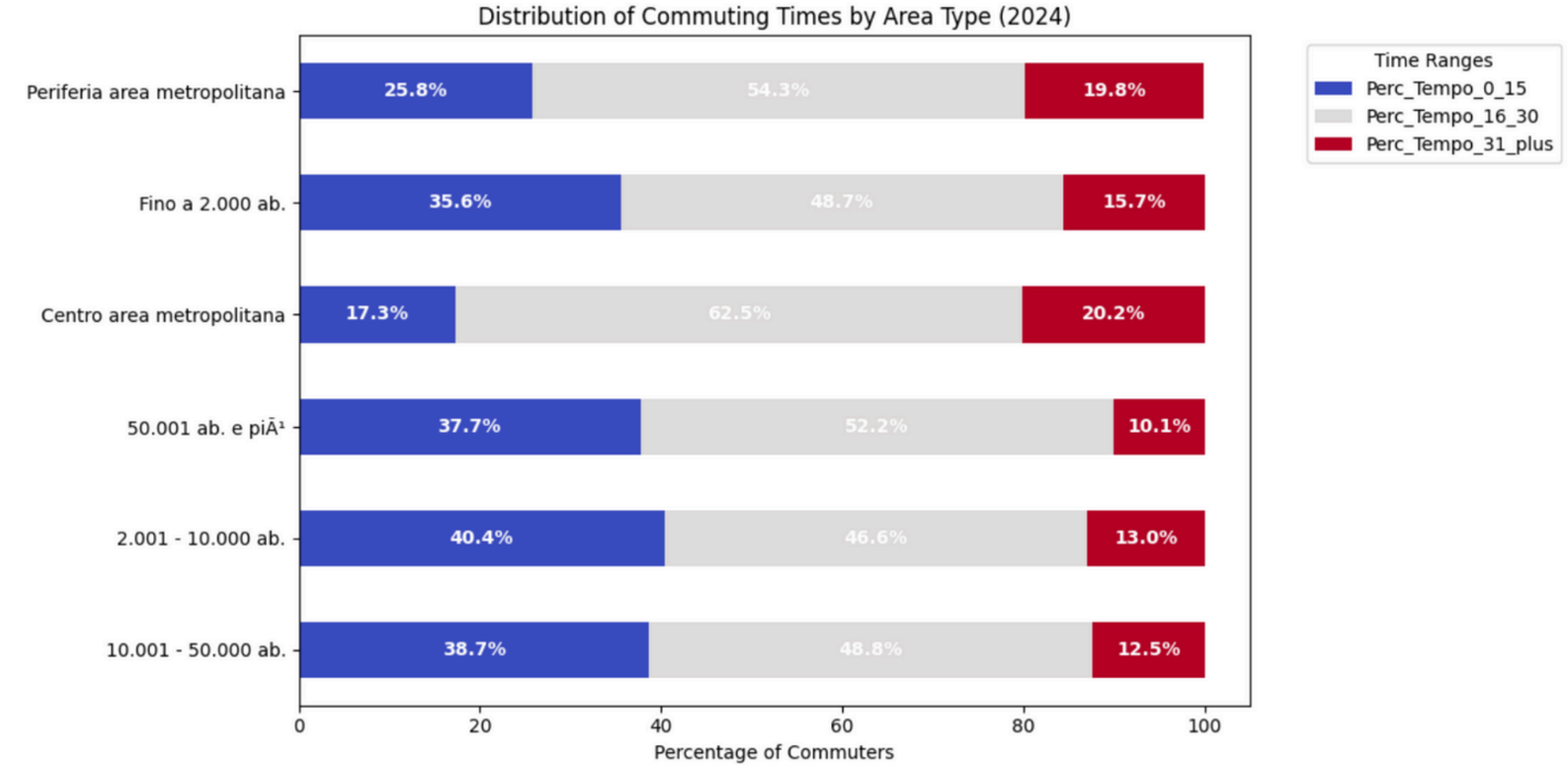
Splitting cleaned data into two main datasets for specialized analysis,:
1. **Urban Analysis** : Based on population size/metropolitan status
2. **Regional Analysis** : Focused on Regions, used for regression analysis including Income

EDA: Temporal Evolution of Means of Transport by Urban Area (2001–2024)





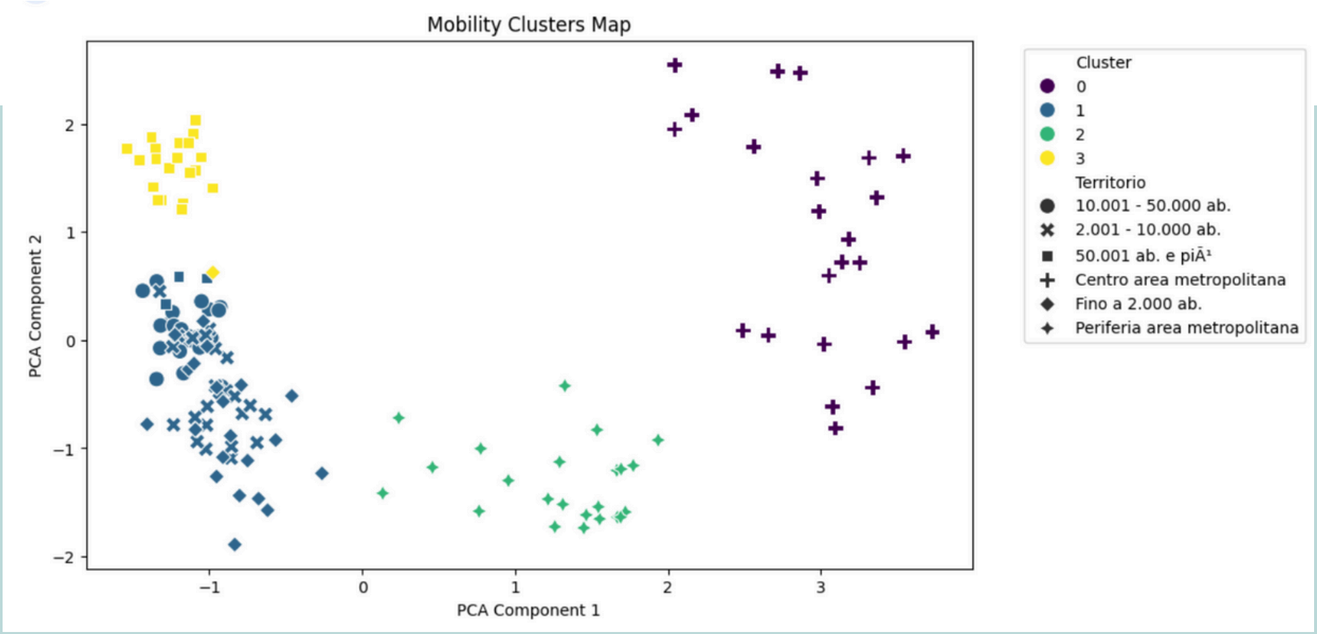
EDA: Commuting Time Distribution by Urban Area (2024)

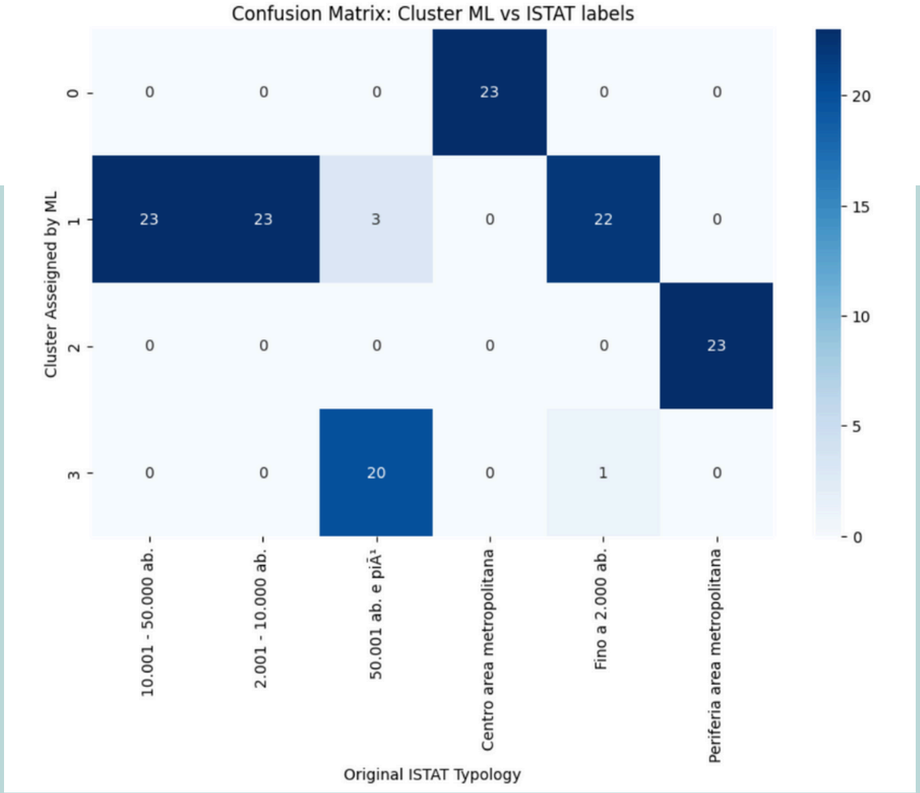


The Metropolitan Area Center has the longest commuting times, with 20.2% of commuters taking over 30 minutes. There are also some area types that share similar results, so it could be good to analyze the possibility of clustering

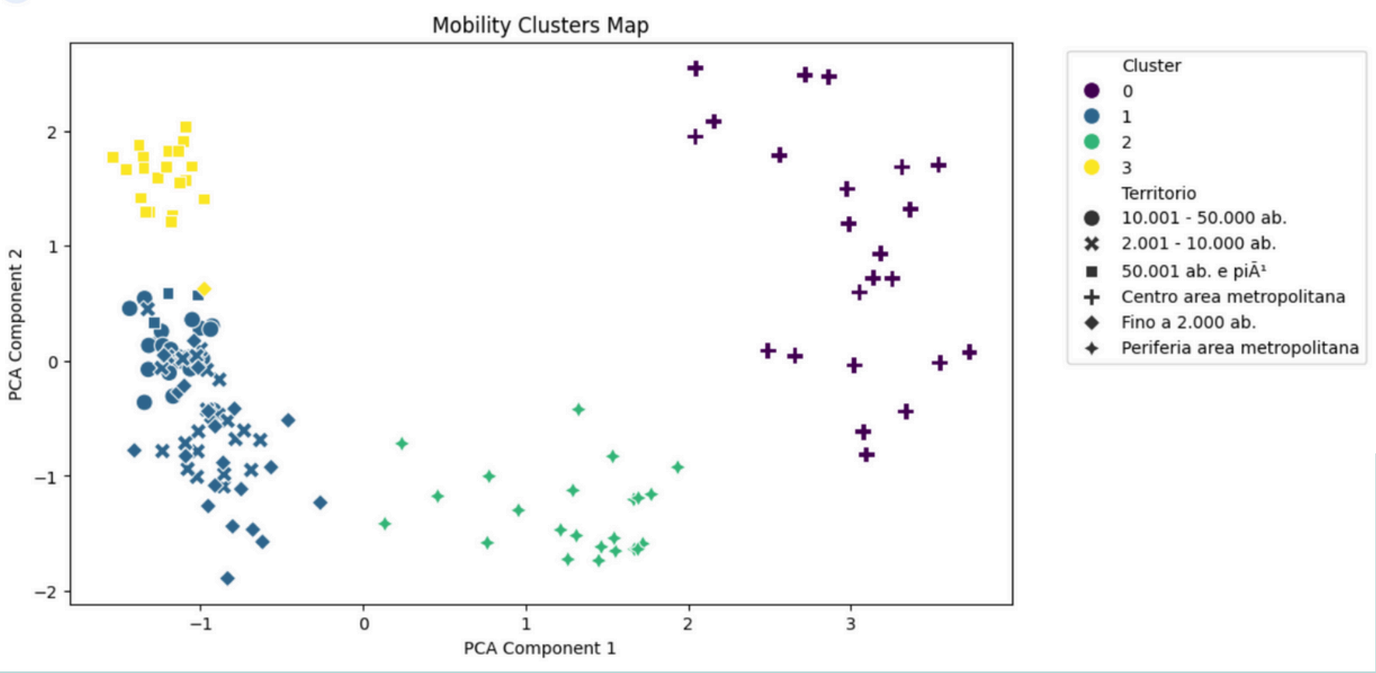
Unsupervised Learning: Clustering Urban Mobility Profiles

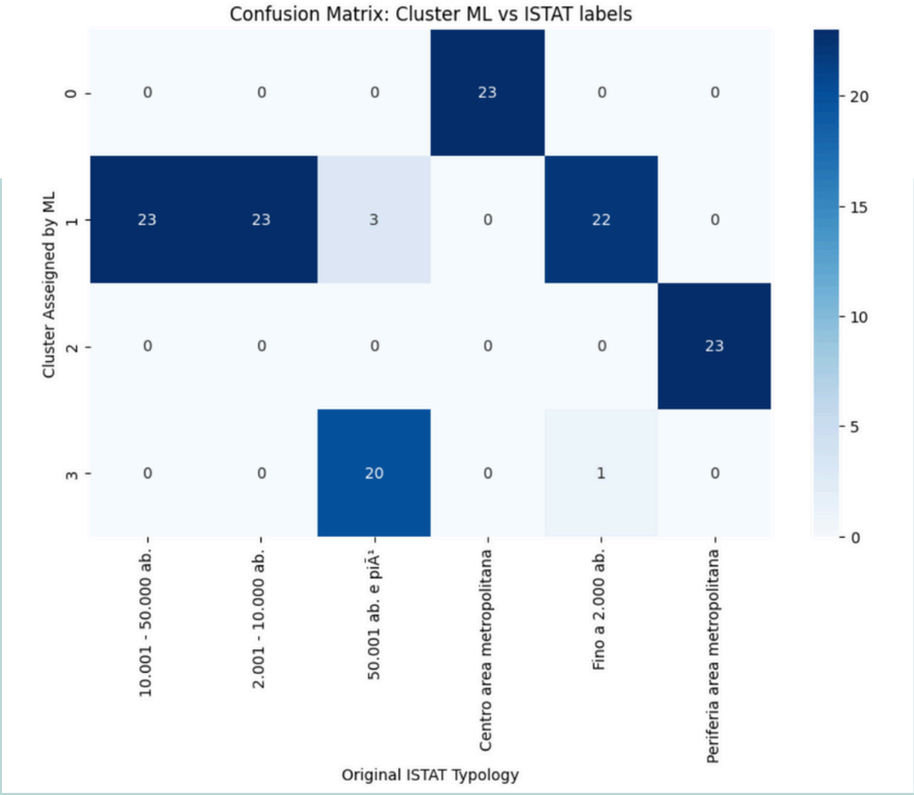
- Objective: Identify natural groupings of urban areas based on mobility behavior rather than ISTAT definitions
- Methodology: K-Means clustering applied to four key features: Car Use, Public Transport Use, Active Mobility, and Long Commuting (Percentages)





Unsupervised Learning: Clustering Urban Mobility Profiles





- Four clusters were determined to be optimal (following the Elbow Method)
- The Machine Learning analysis demonstrated nearly total uniformity in mobility behavior across all urban areas under 50,000 inhabitants, suggesting that the detailed ISTAT distinctions for small municipalities are redundant regarding commuting patterns.

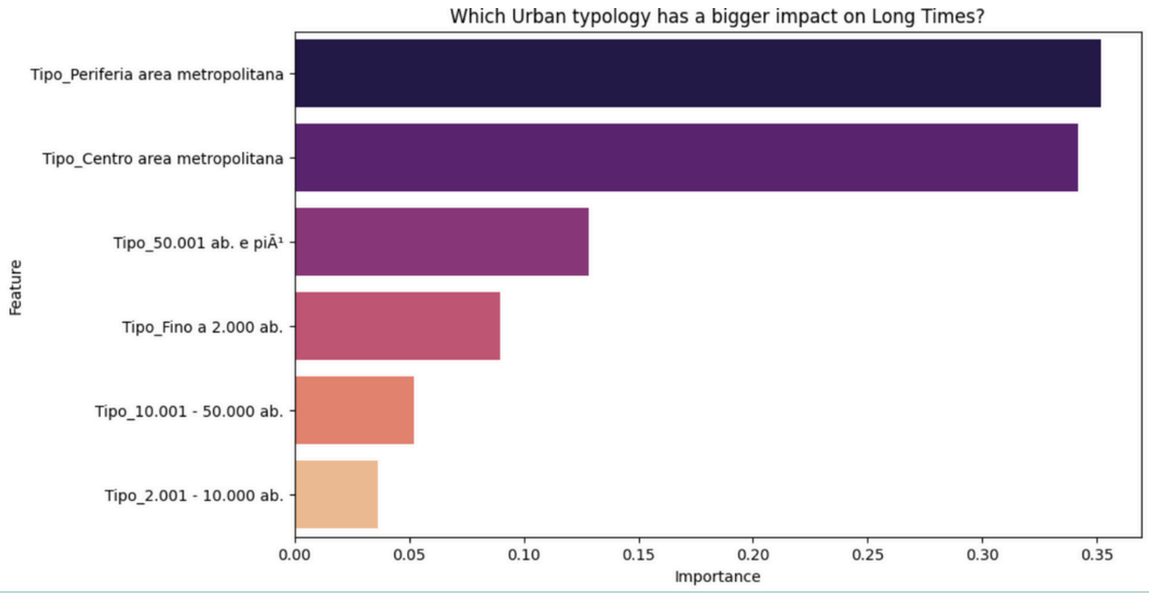
The Impact of Urban Typology: A Random Forest Approach

- **Objective**: Isolate the structural impact of territory on commuting times, stripping away behavioral variables (transport modes). In particular, rank which territories are the strongest predictors for congestion and which ones facilitate the '15-minute city' lifestyle.

- **Methodology**:
- Random Forest Regressor (chosen for non-linearity and feature ranking)
- One-Hot Encoding applied to ISTAT Urban Areas (Center, Periferia, Small Towns, etc.)
- Targets: Long Commuting (>31 min) vs. Proximity (0-15 min)

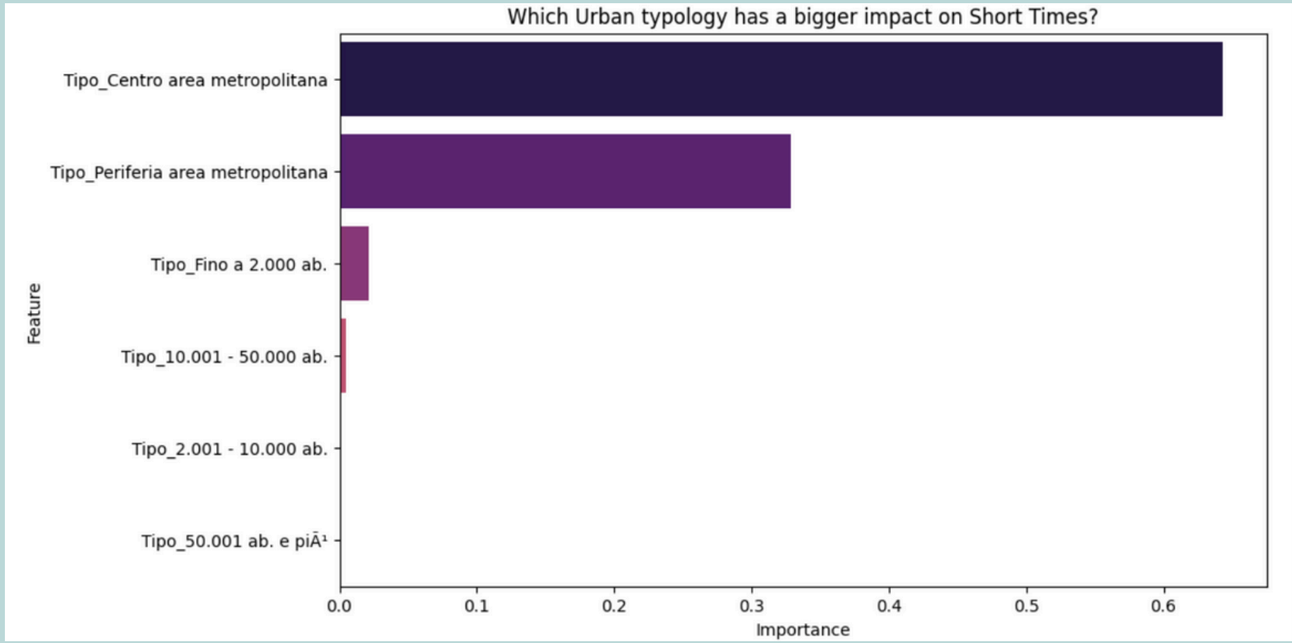


The Impact of Urban Typology: A Random Forest Approach

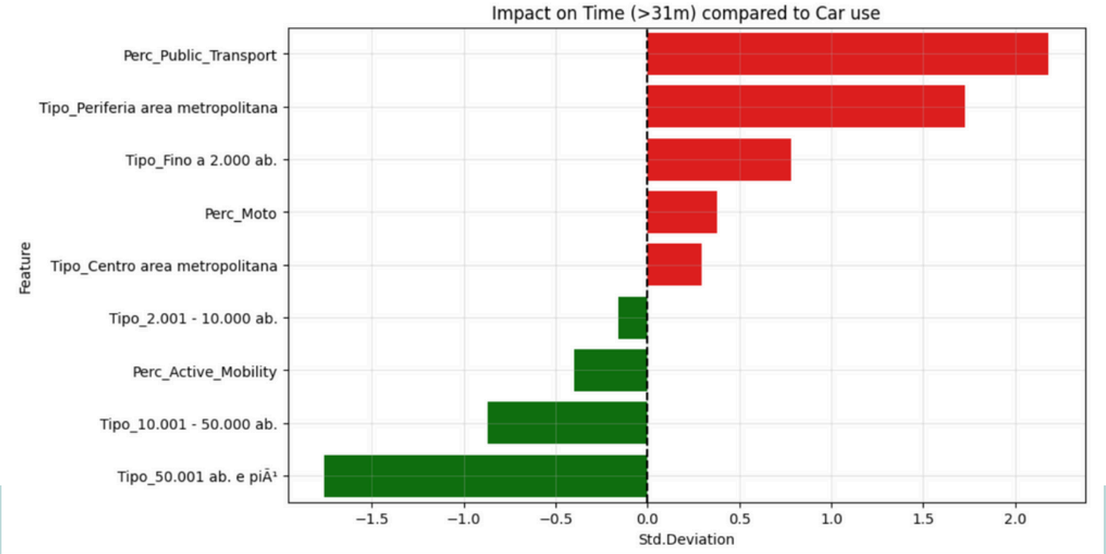


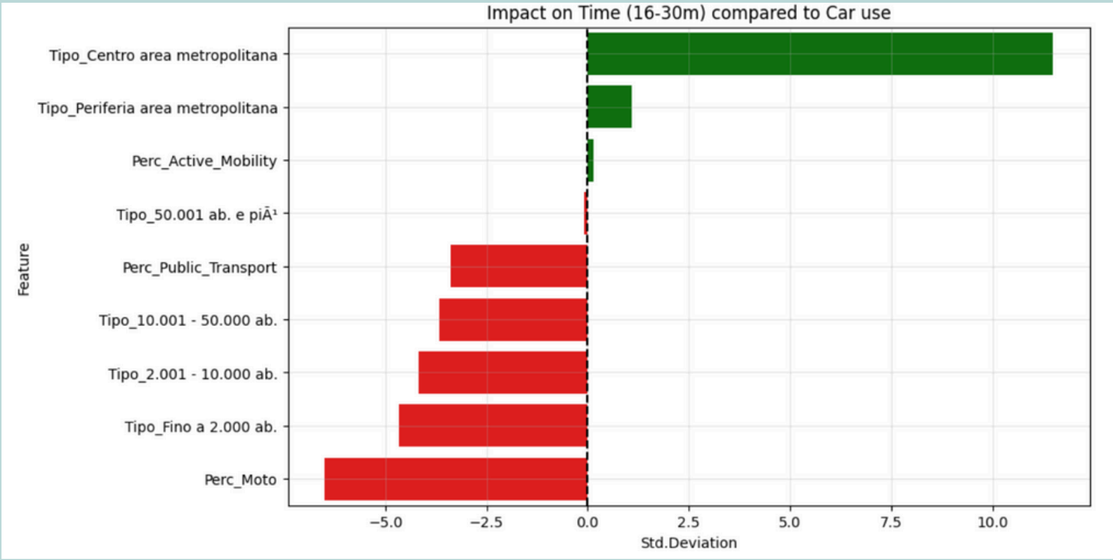
Here we see the hierarchy of structural problems. The Metropolitan Area Periphery is the dominant feature together with the Center. This tells that long commuting times are largely a result of urban planning: if you live in these areas, you face a structural disadvantage regardless of personal efficiency.

When we flip the target to short commutes, we see a pretty interesting result. Here, the Metropolitan Area Center has an enormous impact and it is definitely the strongest predictor for quick trips, followed by the peryphery.



The Impact of Urban Typology: Linear Regression



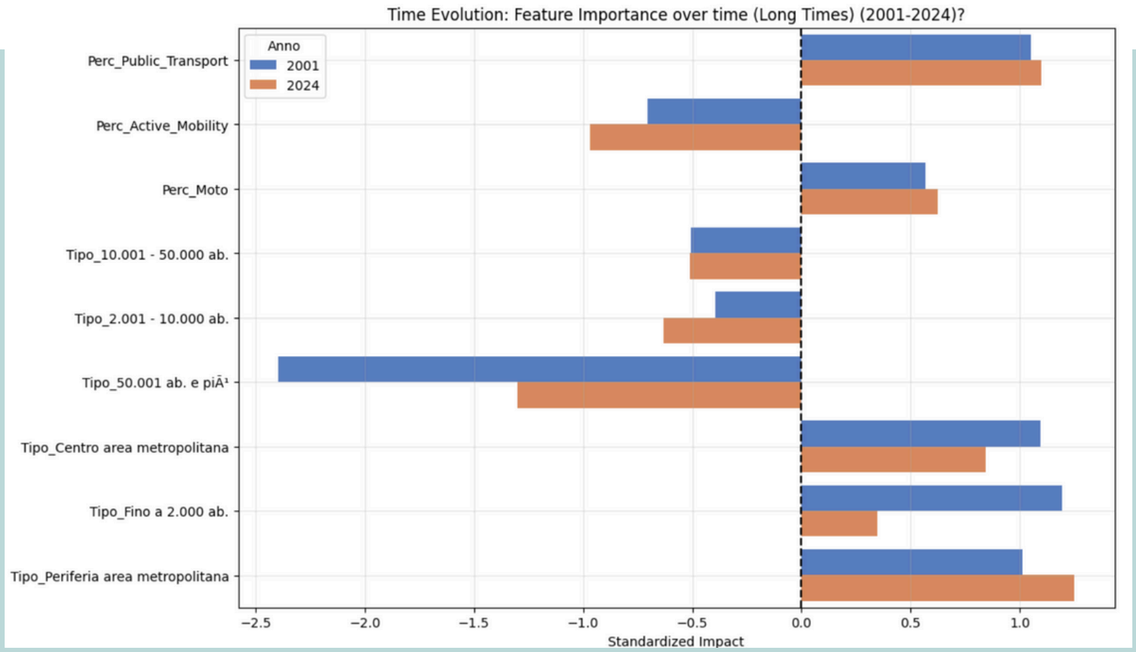


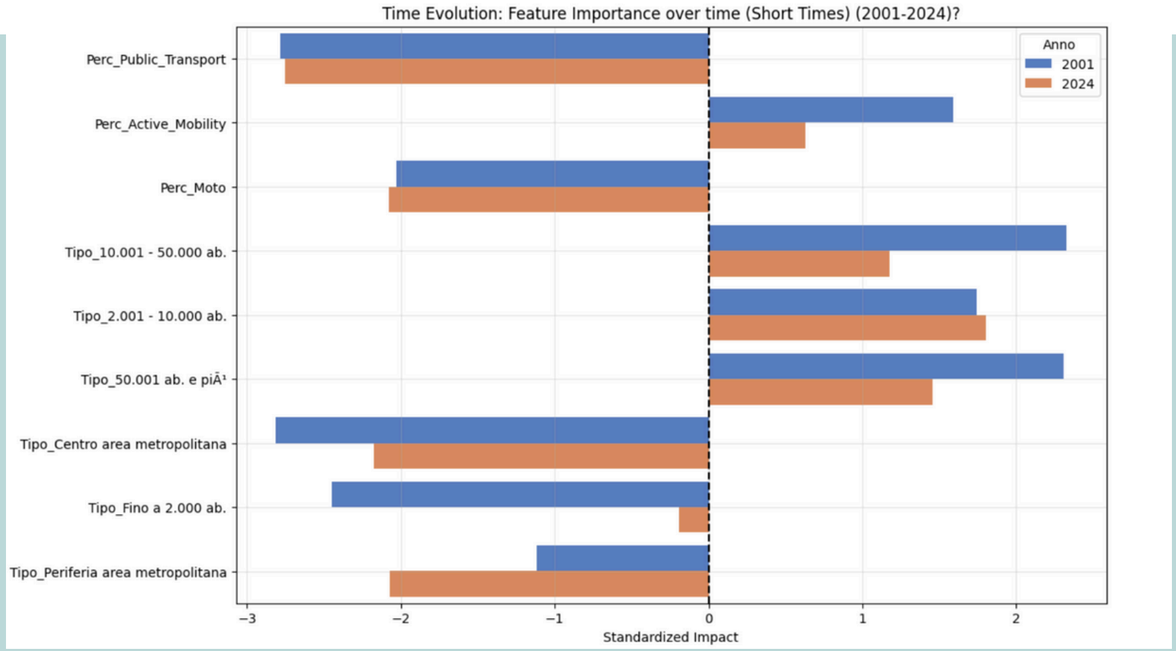
Peforming a Linear Regression integrating also Means of Transport feaures (using Car as a baseline), an important result emerges: for **Long commuting times**, Public Transport becomes even more important than the Periphery Area (which was the strongest with Random Forest without means of transport), while for **Short commuting times** , the results are coherent with the ones observed in the previous analysis.

Also, motorbike emerges as a strongly positive factor to reduce commuting times, probably thanks to its ability to avoid congestion (traffic filtering).

The Impact of Urban Typology over Time

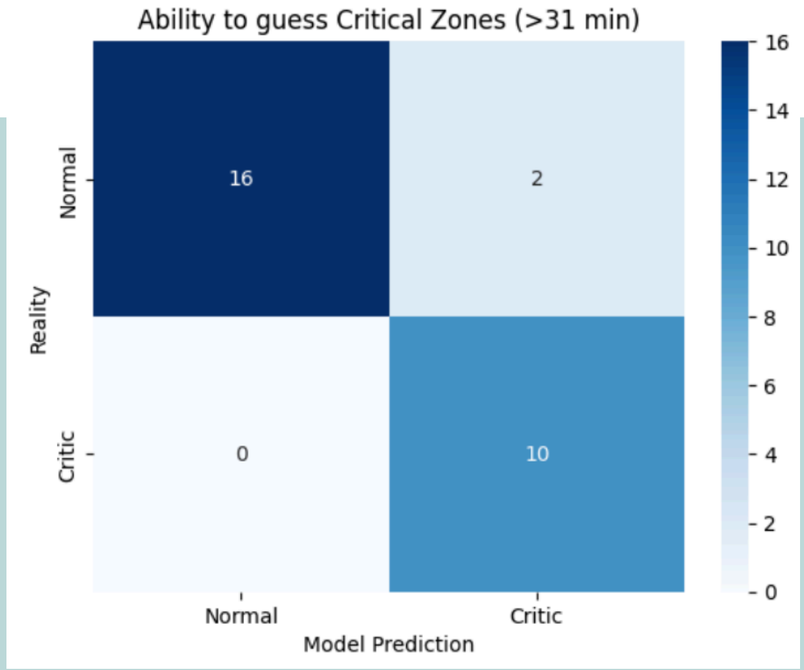
Analyzing the impact of all these features over time, it is possible to see a story of polarization: The **Periphery** has become a worse place for commuters: this is probably due to the fact that the infrastructure in these outer rings has likely failed to keep up with population growth, creating severe bottlenecks. **Small Towns** have become a lot less relevant to the commuting problem, meanwhile, the legendary efficiency of **Medium Towns** is fading.
Talking about **means of transport**, motorbikes and public transport seem to have similar behaviors over time, while active mobility has lost a good part of his 2001 effect.



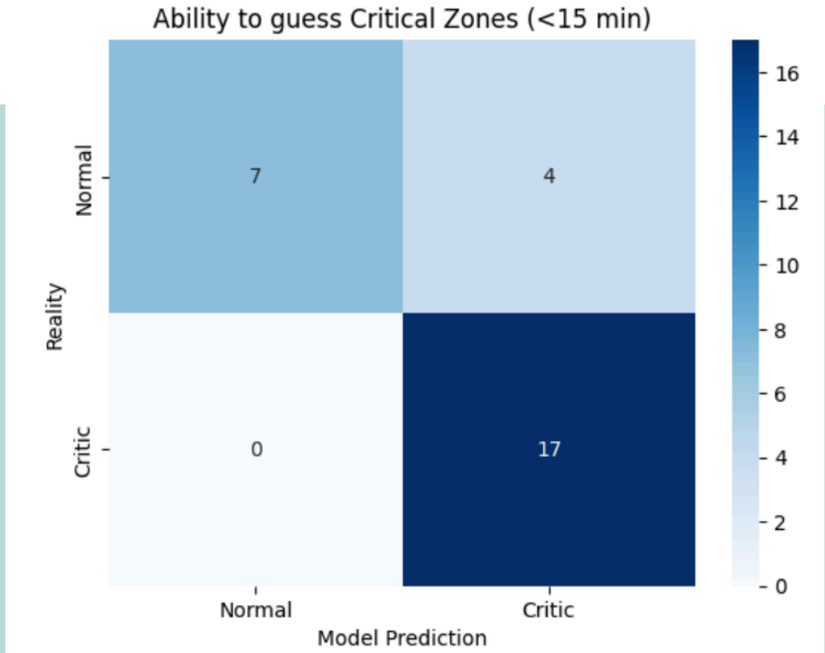


Logistic Regression and Classification: Predicting Critical Zones

- Defined a binary target based on the national median to flag high-risk zones
- Hybrid model combining Urban Typology and Transport Modes
- Prediction of critical areas seems great (91% - 86%), so urban suddivision of territory is valid



Accuracy= 91%



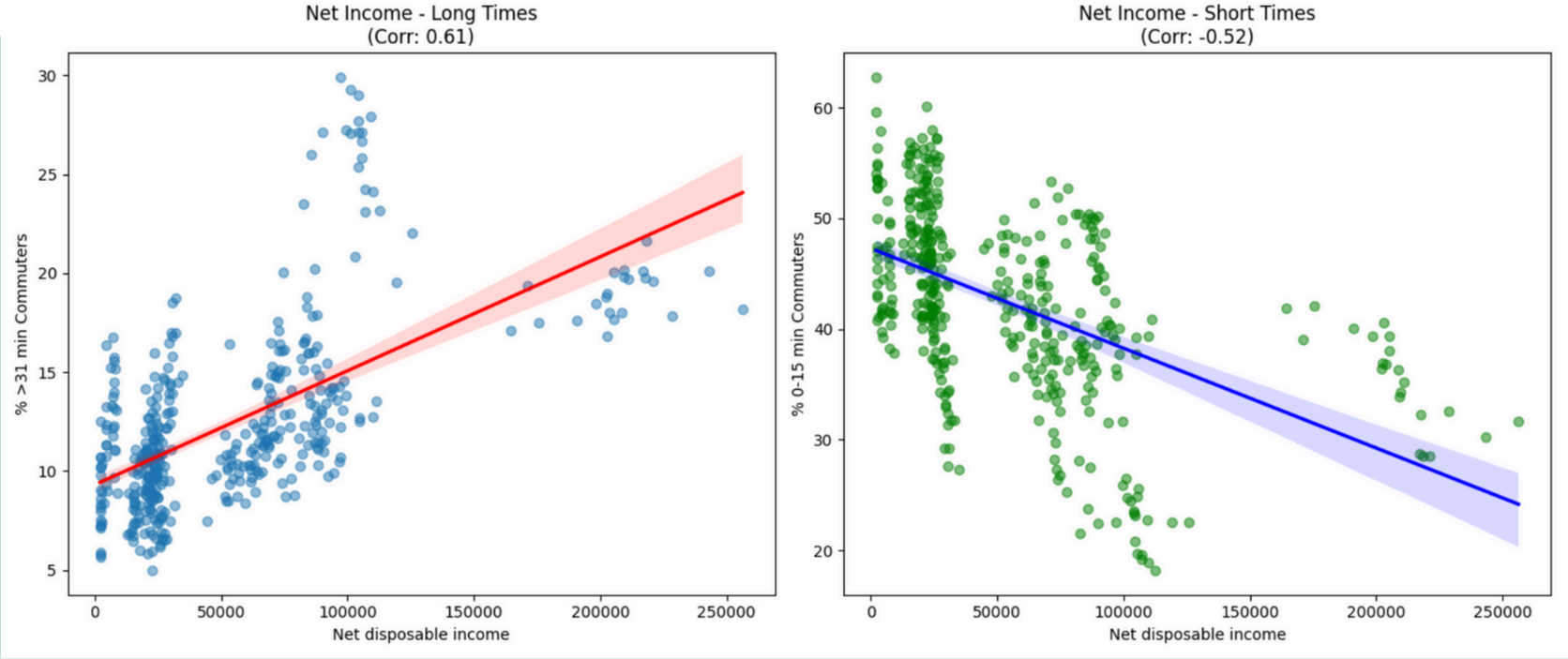
Accuracy=86%

Regional Analysis – Income vs Commuting Times

- Data Integration: Enriched the Regional Dataset by merging mobility stats with Net Disposable Income
- Methodology: Applied Linear Regression to correlate regional wealth with travel duration

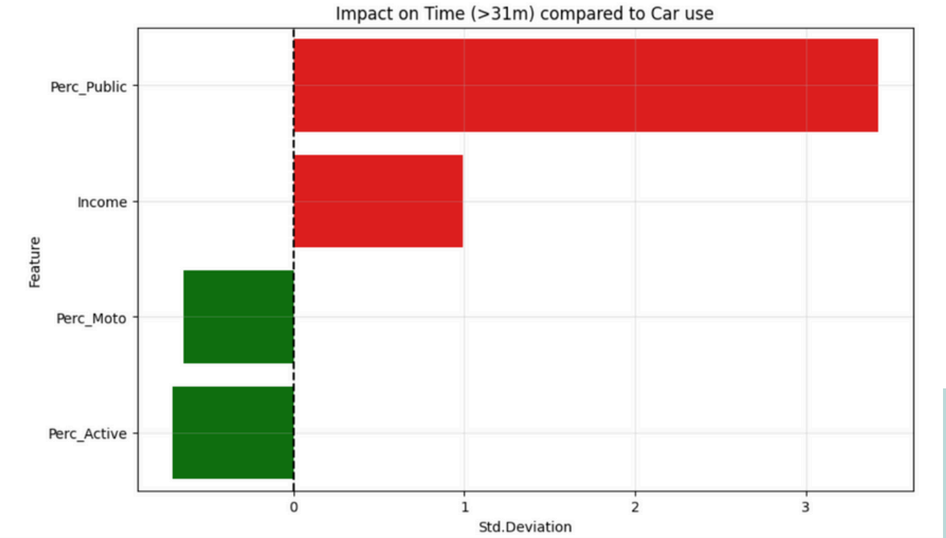
Key Findings:

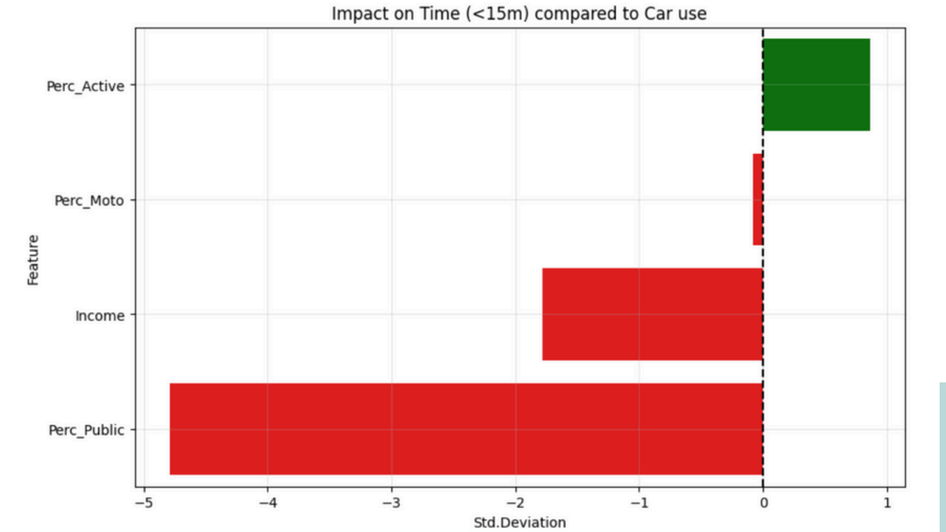
- Long Commutes (>31 min): Wealth often drives suburbanization, leading to longer distances
- Short Commutes (<15 min): Higher disposable income reduces the likelihood of living and working in the same neighborhood



The Impact of Regions and Income: Linear Regression

Model: Standardized Linear Regression on regional data, comparing modal share and income against the percentage of commuters taking >31 minutes. Private Car use is the baseline





- Public Transport (+3.42): shifting from a car to public transport drastically increases the probability of a commute exceeding 30 minutes,. This quantifies the structural time inefficiency (waiting times, transfers) of Italian public transit.
- Income (+0.99): wealth buys distance or better quality housing outside chaotic centers, despite accepting longer travel times.

- Public Transport (-4.79): Public transport shows the strongest negative relationship with short commutes, confirming it is structurally incompatible with trips under 15 minutes
- Income (-1.78): Higher income significantly decreases the probability of having a short commute. This confirms that affluent socioeconomic groups are generally less involved in proximity commuting

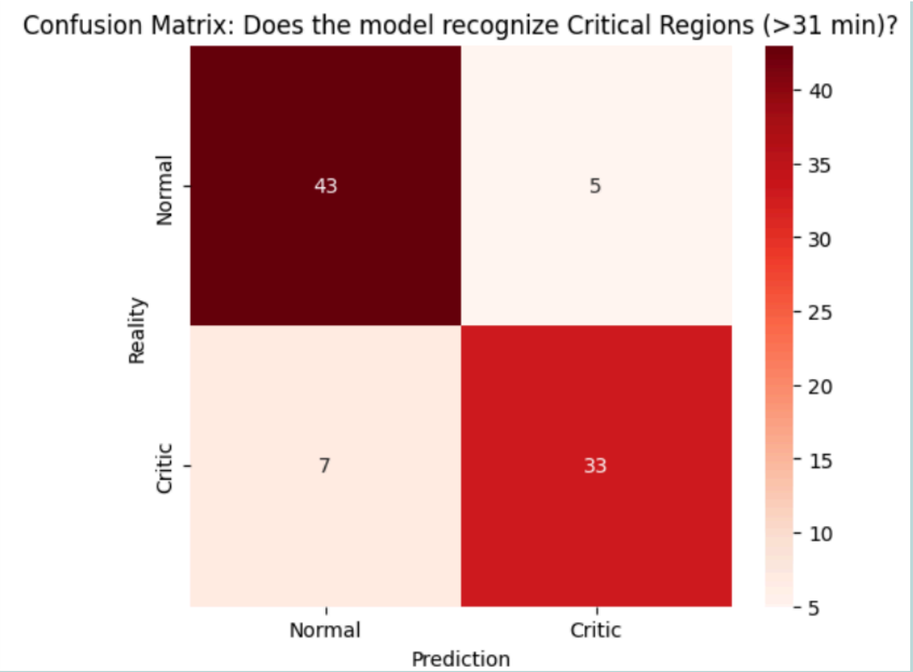
Motorcycles have negative coefficients for both shortand long trips indicate that motorcycle usage peaks in the medium range (16-30 min).

- High setup cost (helmet, gear) makes bikes impractical for very short distances compared to walking or driving.
- Motorcycles are efficiency tools. Thanks to traffic filtering, they compress what would be a "long" car commute (e.g., 40 min) into a "medium" ride (~25 min).

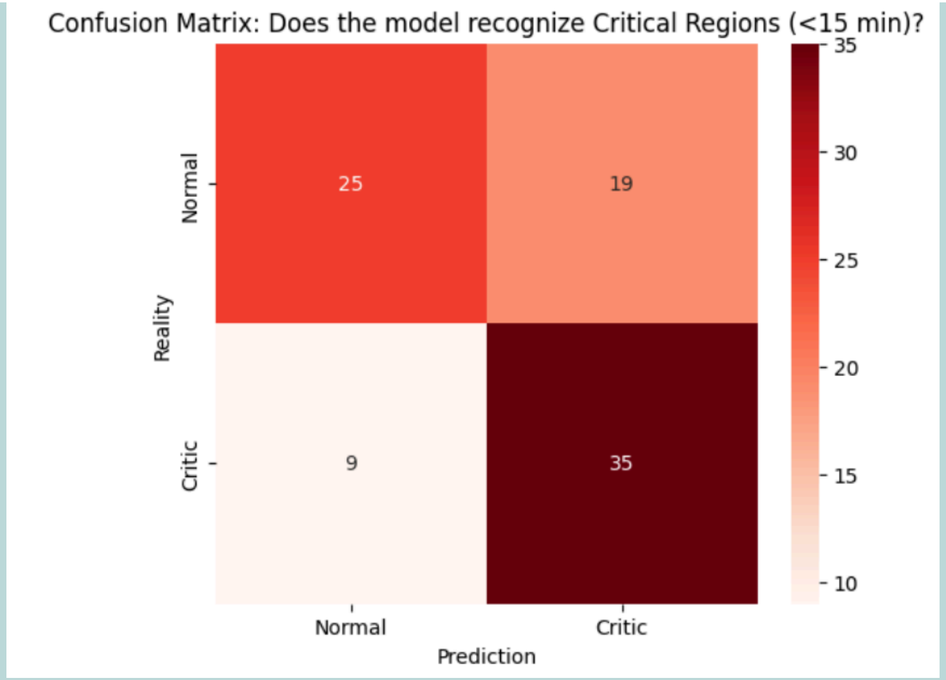
Logistic Regression and Classification: Predicting Critical Regions

- Only Regional dummy variables (Transport mode and Income excluded) to better isolate pure territorial risk
- model predicts critical long commuting times with 86% accuracy, but drops to 68% for short times.

Delays are structural: they depend on regional assets like highway congestion or rail efficiency.



Proximity is a micro-urban phenomenon—living 10 minutes from work is a matter of local neighborhood design, making the Region variable less predictive.



Resources and Links

- Complete Notebook (Google Colab)
- ISTAT Dataset 1 (Commuting Time for Region and Territory)
- ISTAT Dataset 2 (Disposable Income of Families in Italian Regions)

Thank You for your Attention!

Any questions?